# TARGET : JEE (M + A) : 2023

# ENTHUSIAST SRG TEST - 07

**TEST TYPE: OFFLINE** **DATE : 18.09.2022** **PATTERN : JEE ADVANCED**

**Time : 3 Hours** **Maximum Marks : 210**

## SOLUTION

## PART-1 : PHYSICS

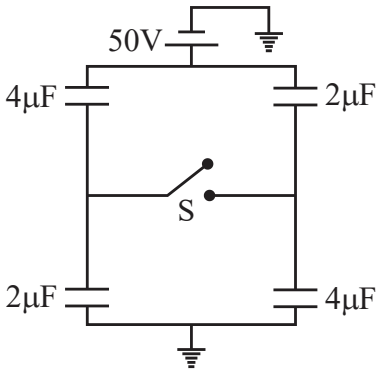
### SECTION–I(i) : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

*Full Marks* : +3 If only the bubble corresponding to the correct option is darkened.
*Zero Marks* : 0 If none of the bubbles is darkened.
*Negative Marks* : –1 In all other cases

---

**1.** The circuit was in the shown state for a long time. Now if the switch S is closed then the net charge that flows through the switch S, will be

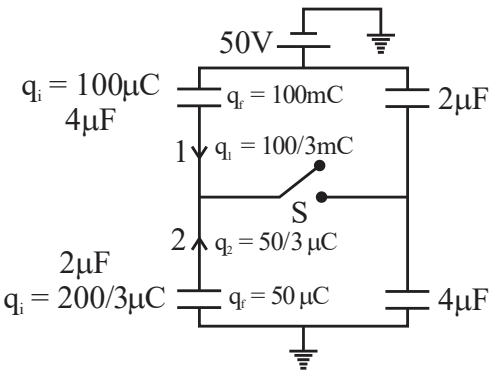



(A) $\frac{400}{3}\mu C$ (B) $100\,\mu C$

(C) $\frac{100}{3}\mu C$ (D) $50\,\mu C$

**Ans.** D

**Sol.** Initial and final charges are marked on $4\,\mu F$ and $2\,\mu F$ capacitors as shown.

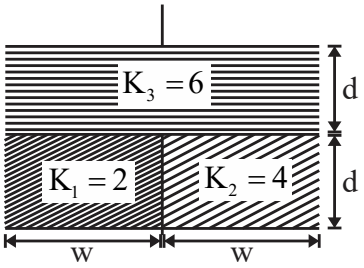


Hence charge passing through segment 1 and 2 are

$$q_1 = \frac{100}{3}\mu C \qquad q_2 = \frac{50}{3}\mu C$$

$\therefore$ charge through switch = $q_1 + q_2 = 50\,\mu C$.

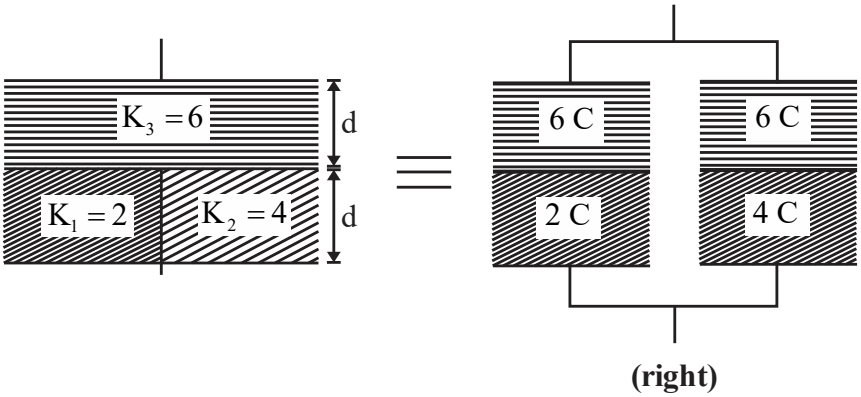
**2.** A parallel plate capacitor of capacitance C (without dielectrics) is filled by dielectric slabs as shown in figure. Then the new capacitance of the capacitor is:




(A) 3.9 C (B) 4 C
(C) 2.4 C (D) 3 C

**Ans.** A

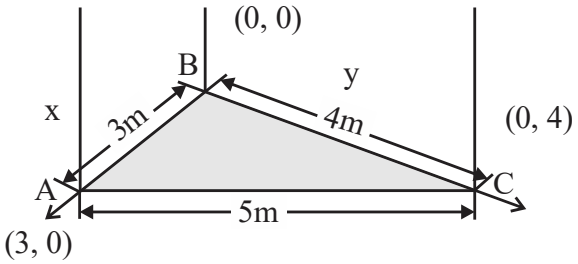
**Sol.**




**(right)**

The equivalent capacitance $C_{eq} = \frac{2C \times 6C}{2C + 6C} + \frac{6C \times 4C}{6C + 4C} = 3.9C$

**3.** A uniform right-angled triangular lamina is suspended from the ceiling with the help of three light strings. The plane of the lamina is horizontal and the strings are vertical as shown in the figure. In which of the following options, tensile forces $T_A$, $T_B$ and $T_C$ developed in the strings connected to the vertices A, B and C respectively of the lamina are arranged in correct order of their magnitudes?




(A) $T_A > T_B > T_C$ (B) $T_B > T_A > T_C$
(C) $T_C > T_A > T_B$ (D) $T_A = T_B = T_C$

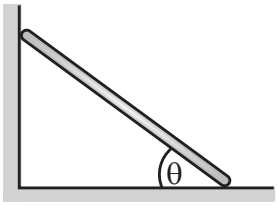
**Ans.** D

**Sol.** Torque about BC

$$T_A \times 3 = w \times 1 \qquad T_A = \frac{w}{3}$$

$$T_B = \frac{w}{3} \qquad T_C = \frac{w}{3}$$

**4.** A uniform steel rod leans against a frictionless wall in static equilibrium. Frictional force between the lower end and the floor is less than its limiting value by a finite amount. The rod is supplied some amount of heat so that it expands. Assume that the coefficient of friction does not change on heating. What can you conclude regarding contact forces from the wall and from the floor?




(A) From the wall it decreases and from the floor it increases.
(B) From the wall it decreases and from the floor it decreases.
(C) From the wall as well as from the floor it increases.
(D) From the wall as well as from the floor it decreases.

**Ans.** D

**Sol.** Angle $\theta$ will increase, so both contact force will increases.

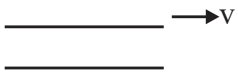
**5.** A parallel plate capacitor consists of square plates of edge length a separated by a distance d that is much smaller that the edge length. It is charged to a potential V and made to move with a constant velocity v directed along one of its edges. How much magnetic field exists inside the capacitor?

(A) $\frac{\mu_0 \varepsilon_0 v V}{d}$ (B) $\frac{\mu_0 \varepsilon_0 v V}{a}$

(C) $\frac{\mu_0 \varepsilon_0 dv V}{a^2}$ (D) None of these

**Ans.** A

**Sol.** $\frac{d\tau}{dx} = \sigma v \qquad B = \mu_0 \sigma v \qquad \frac{\sigma}{\varepsilon_0} d = V$




$$B = \frac{\mu_0 v V \varepsilon_0}{d}$$

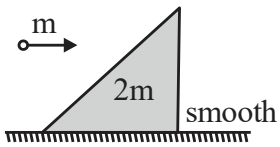
**SECTION–I(ii) : (Maximum Marks : 40)**

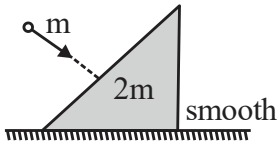
- This section contains **TEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

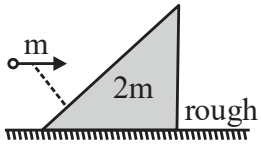
| | | |
|---|---|---|
| *Full Marks* | : +4 | If only (all) the correct option(s) is (are) chosen. |
| *Partial Marks* | : +3 | If all the four options are correct but ONLY three options are chosen. |
| *Partial Marks* | : +2 | If three or more options are correct but ONLY two options are chosen, both of which are correct options. |
| *Partial Marks* | : +1 | If two or more options are correct but ONLY one option is chosen and it is a correct option. |
| *Zero Marks* | : 0 | If none of the options is chosen (i.e. the question is unanswered). |
| *Negative Marks* | : –2 | In all other cases. |

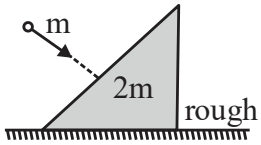
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

**6.** In which of the following system(s) linear momentum cannot be conserved during collision along horizontal line (parallel to base) ?

(A) 


(B) 


(C) 


(D) 


**Ans.** CD

**Sol.** If ground is rough, impulse will be non-zero.

**7.** A particle moving with kinetic energy = 3J makes an elastic head-on collision with a stationary particle which has twice its mass. During the impact :

(A) The minimum kinetic energy of the system is 1J.

(B) The maximum elastic potential energy of the system is 2J.

(C) Momentum and total energy are conserved at every instant.

(D) All of the above are correct

**Ans.** ABCD

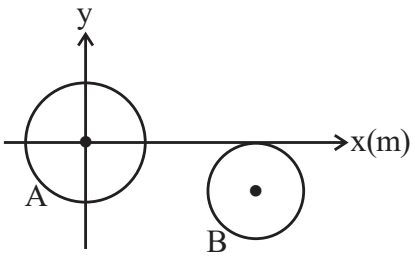
**Sol.** $\frac{1}{2}mv^2 = 3J$




$KE_{sys-cm} = \frac{1}{2} \times \frac{2m}{3} \times v^2 = \frac{mv^2}{3} = 2J$

$\min KE \Rightarrow 2J$ max pot energy = 2J

**8.** Two smooth balls A and B, each of mass m and radius R, have their centres at (0, 0, R) and at (5R, –R, R) respectively, in a coordinate system as shown. Ball A, moving along positive x-axis, collides with ball B. Just before the collision, speed of ball A is 4 m/s and ball B is stationary. The collision between the balls is elastic.




(A) Velocity of the ball A just after the collision is $\left(\hat{i}+\sqrt{3}\,\hat{j}\right)$ m/s

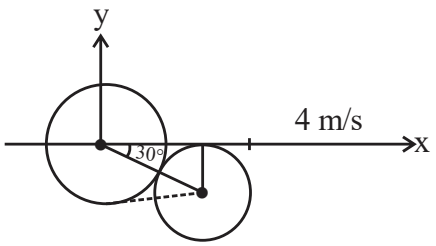
(B) Velocity of B just after collision is zero.

(C) Impulse of the force exerted by A on B during the collision, is equal to $\left(3m\hat{i}-\sqrt{3}m\hat{j}\right)$kg – m/s

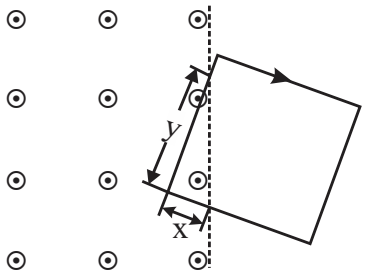
(D) Impulse of the force exerted by A on B is $\left(4m\hat{i}\right)$kg – m/s

**Ans.** AC

**Sol.** Velocity wall exchange along the line of impact.




**9.** A rigid square loop of side length $l$ carrying an electric current is held motionless on a frictionless horizontal tabletop. A uniform magnetic field pointing upwards is switched on everywhere to the left of the dashed line as shown in the figure and then the loop is released. Considering different lengths of the side segments x and y (x < y), which of the following conclusions can you make?




(A) If $x<\frac{l}{2}$ and $y=\frac{l}{2}$, the loop starts rotating anticlockwise

(B) If $x<\frac{l}{2}$ and $y<\frac{l}{2}$, the loop starts rotating anticlockwise

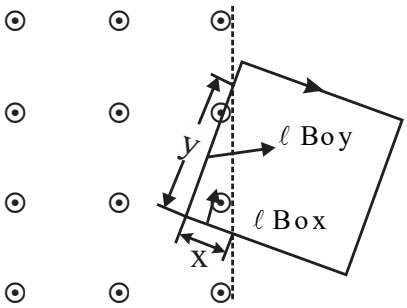
(C) If $x<\frac{l}{2}$ and $y>\frac{l}{2}$, the loop starts rotating clockwise

(D) If $x<\frac{l}{2}$ and $y>\frac{l}{2}$, more information is required to decide which way the loop starts rotating

**Ans.** ABD

**Sol.** If $x + y < l$,

loop rotates acω



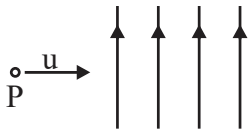

**10.** A positively charged particle P crosses a region of uniform electric field of intensity E. Velocity u of the particle at the entry on the left side of the region is perpendicular to the electric field as shown in the first figure. Speed of the particle at exit on the right side is $v_1$. Now the experiment is repeated with a uniform magnetic field of induction B superimposed on the electric field as shown in the second figure, keeping entry velocity of the particle unchanged. In this experiment, speed of the particle at exit on the right side is $v_2$. What can you conclude regarding the speeds $v_1$ and $v_2$?

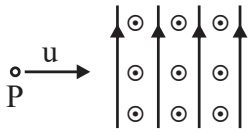
(A) If $u < \frac{E}{B}$, then $v_2 < v_1$

(B) If $u = \frac{E}{B}$, then $v_2 < v_1$

(C) If $u = \frac{E}{B}$, then $v_2 = v_1$

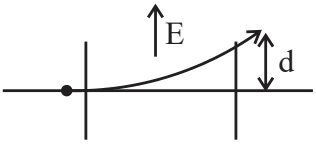
(D) If $u > \frac{E}{B}$, then $v_2 < v_1$




**Ans.** ABD

**Sol.** $qEd = \Delta KE$, more d more change m speed.




**11.** A human body has a surface area of approximately $1m^2$. The normal body temperature is 10 K above the surrounding room temperature $T_0$. Take the room temperature to be $T_0 = 300$ K. For $T_0 = 300$ K, the value of $\sigma T_0^4 = 460\ Wm^{-2}$ (where $\sigma$ is the Stefan-Boltzmann constant). Which of the following options is/are correct?

(A) The amount of energy radiated by the body in 1 second is close to 60 Joules.

(B) If the surrounding temperature reduces by a small amount $\Delta T_0 << T_0$, then to maintain the same body temperature the same (living) human being needs to radiate $\Delta Q = 4\sigma T_0^3 \Delta T_0$ more energy per unit time

(C) Reducing the exposed surface area of the body (e.g. by curling up) allows humans to maintain the same body temperature while reducing the energy lost by radiation.

(D) If the body temperature rises significantly then the peak in the spectrum of electromagnetic radiation emitted by the body would shift to longer wavelengths.

**Ans.** BC

**12.** You are a consultant for a cookware manufacturer who wishes to make a pan that will have two features:

(1) Absorb thermal energy from a flame as quickly as possible, and

(2) Have an inner surface that remains as hot as possible when cooking.

Consider only radiation. You should recommend a pan with the

(A) Outer and inner surfaces black.

(B) Outer and inner surfaces shiny.

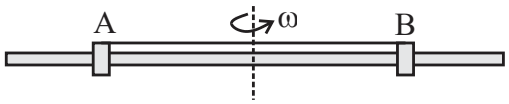
(C) Outer surface shiny and inner surface black.

(D) Outer surface black and inner surface shiny.

**Ans.** D

**Sol.** Outer surface should absorb max and inner surface should radiate less.

**13.** Two identical small beads A and B threaded on an almost inertia-less horizontal rod of length $2l$ are connected by a light inextensible cord of length $1.2l$. Initially the beads are adjusted equidistant from the centre of the rod and the rod is rotating about its central vertical axis at an angular velocity 1/3 rad/s. At some point of time, the cord is cut and the beads begin to slide on the rod without friction. Neglect effects of gravity. Which of the following statements are correct?




(A) Path of a bead in an inertial frame is a spiral.

(B) Path of a bead in an inertial frame is a straight line.

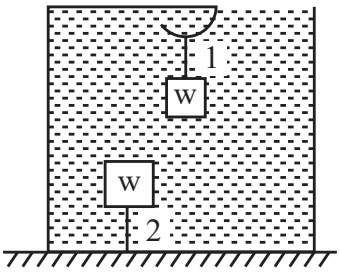
(C) To leave the rod, the beads take 4s after the cord is cut.

(D) To leave the rod, the beads take more than 4s after the cord is cut.

**Ans.** BC

**Sol.** There is no external force so path will be a straight line.

**14.** An iron block and a wooden block are positioned in a vessel containing water as shown in the figure. The iron block (1) hangs from a massless string with a rigid support from the top while the wooden block (2) floats being tied to the bottom through a massless string. If now the vessel starts acceleration towards right.




(A) Iron block gets deflected towards right.

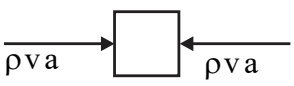
(B) Wooden block gets deflected towards right.

(C) Iron block gets deflected towards left.

(D) Wooden block gets deflected towards left.

**Ans.** BC

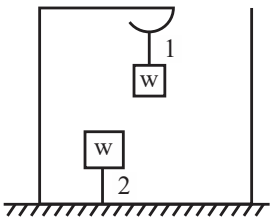
**Sol.** Far from



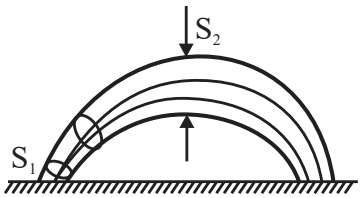

$(F_b)$ pseudo for v

$\sigma > \rho$

so deflect left



**15.** Water jet is projected at an angle to the horizontal. At the point of projection, the area of the jet is $S_1$ and at the highest point, the area of the jet is $S_2$. The initial velocity of projection is u.




(A) The angle of projection is $\cos^{-1}\left(\frac{S_1}{S_2}\right)$

(B) The range on the level ground is $\frac{2u^2}{g}\frac{S_1}{S_2}\sqrt{1-\frac{S_1^2}{S_2^2}}$

(C) The maximum height reached from the ground is $\frac{2u^2}{g}\left(1-\frac{S_1^2}{S_2^2}\right)$

(D) The rate of volume flow is $S_2u$

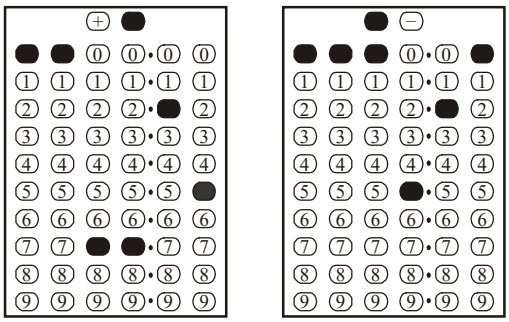
**Ans.** AB

**Sol.** Equation of continuity

$S_1v = S_2v\cos\alpha$

## SECTION-II : (Maximum Marks: 15)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
  **For Example :** If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
  *Full Marks* : +3 If ONLY the correct numerical value is entered as answer.
  *Zero Marks* : 0 In all other cases.

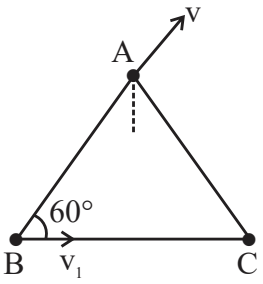
---

**1.** Three small balls A, B and C each of mass m are connected by three thin light rods each of length *l* to make an equilateral triangular structure. The structure is sliding on a frictionless horizontal floor with all the three balls on the floor. At an instant, the ball A is observed moving with a velocity v along the rod AB and the ball B moving along the rod BC. Find moduli of tensile forces developed in rod AB [in N] [Given: m = 1 kg, v = 2 m/s, *l* = 2m]

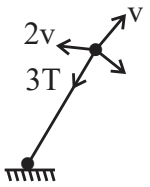
**Ans.** 2

**Sol.** $v_1 \cos 60 = v$

$v_1 = 2v$

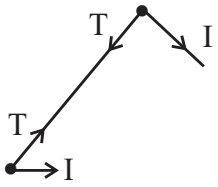



w.r.t ball B

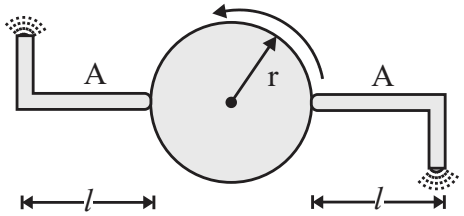



$$3T = \frac{m \times \left(2v \times \frac{\sqrt{3}}{2}\right)^2}{\ell}$$

$$3T = \frac{m3v^2}{\ell} \qquad T = \frac{mv^2}{\ell}$$



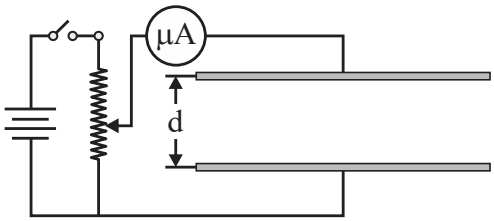

**2.** Shows the top view of a cylindrical can mounted on a turntable. The can is filled with water. At a depth h below the water surface are two horizontal tubes of length $l$ and cross-sectional area a, with right-angle bends at their ends. As the water jets emerge from the tubes, there is a torque $\tau$ exerted on the system given by the expression $t = x\rho gh(r + l)a$, where $\rho$ is the density of the water. Write 'x'.




**Ans.** 4

**Sol.** $C = \underbrace{\rho A v^2}_{\text{thrust force}} \times \underbrace{(2r + 2\ell)}_{\text{length}} \Rightarrow$

**3.** Consider the arrangement shown, where distance between the plates of a parallel plate capacitor is d. When a potential difference V is applied between the plates, electrons are emitted from the negative plate with negligible initial velocity. These electrons are attracted towards the positive plate creating a small electric current, which is registered in a micro-ammeter. What minimum magnetic field must be applied perpendicular to the electric field in the capacitor so that the micro-ammeter registers no current? Denote charge and mass of an electron by symbols e and m respectively and neglect effects of gravity. If minimum magnetic field $\frac{\mu}{2}\frac{1}{d}\sqrt{\frac{nmv}{e}}$. Find 'n'




**Ans.** 8

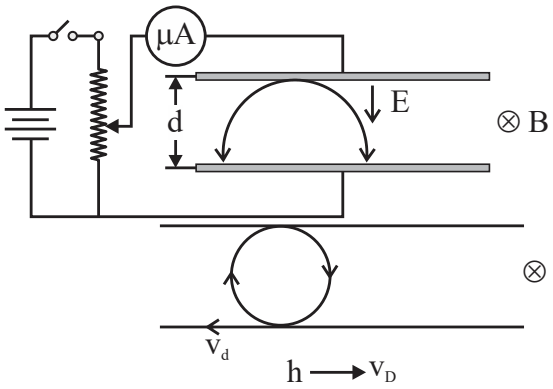
**Sol.** Particle goes in cycloid path

$$R = \frac{mv_0}{qB}$$

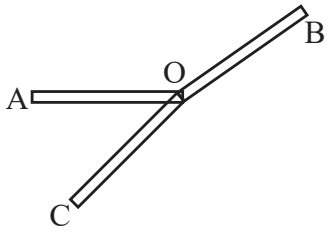
$$R = \frac{m}{qB}\left(\frac{E}{B}\right)$$

$$\frac{d}{2} = \frac{m}{eB^2} \times \frac{V}{d}$$

$$\boxed{B = \sqrt{\frac{2mv}{ed^2}}}$$




**4.** Three rods AO, BO and CO having equal lengths and equal area of cross section are connected as shown in the figure. The temperature of ends A, B and C are 30°C, 60°C and 90° respectively. The thermal conductivities of rods AO, BO and CO are in the ratio of 1 : 2 : 3. What is the temperature of junction O? [in °C]




**Ans.** 70

**Sol.** $\frac{T_0 - T_A}{R_1} = \frac{T_0 - T_C}{R_2} = \frac{T_0 - T_B}{R_3}$

**5.** Two identical plates of metal are welded end to end as shown in figure (A), 20 cal of heat flows through it in 4 minutes when the ends were maintained at different temperatures. If the plates are welded as shown in figure (B) and the ends were maintain at same temperature as in first case, the same amount of heat will flow through the plates in how many minutes ?



**(A)** **(B)**

**Ans.** 1

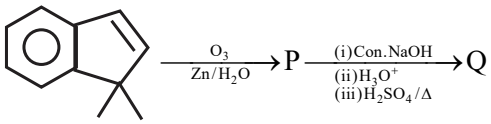
# PART-2 : CHEMISTRY

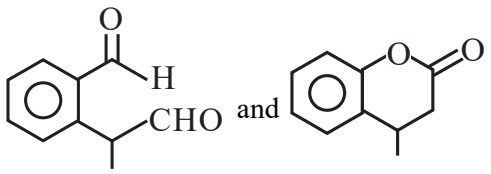
## SECTION–I(i) : (Maximum Marks : 15)

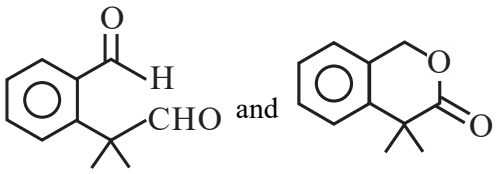
- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

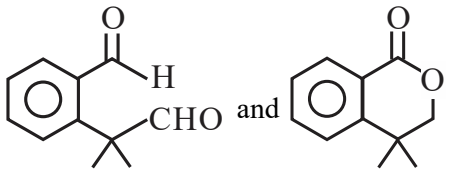
*Full Marks* : +3 If only the bubble corresponding to the correct option is darkened.
*Zero Marks* : 0 If none of the bubbles is darkened.
*Negative Marks* : –1 In all other cases

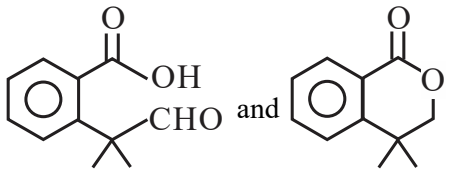
**1.** Identify P and Q respectively



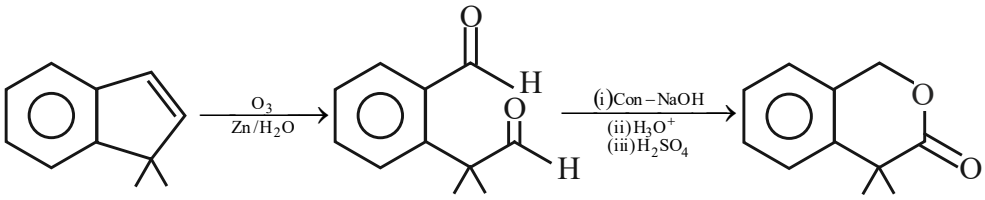

(A)  and

(B)  and

(C)  and

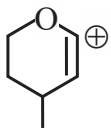
(D)  and

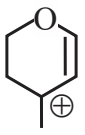
**Ans.** B

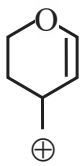
**Sol.** 

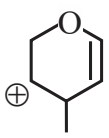

(P) (Q)

**2.** The most stable carbocation is

(A) 

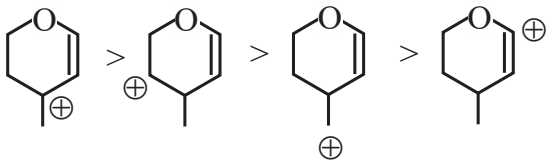
(B) 

(C) 

(D) 

**Ans.** B

**Sol.** Order of stability



**3.** Calculate the electronegativity of X (Bond energy ofAX bond is 78 KCal/mole). Given that bond energy of $A_2$ and $X_2$ are 36 and 49 KCal/mol respectively. Given that Electronegativity of A is 3

(A) 2.752 (B) 1.752

(C) 3.752 (D) 0.752

**Ans.** B

**Sol.** $X_A - S_X = 0.208\sqrt{\Delta_{AX}}$

$\Delta_{A-X} = B.E_{AX} - \sqrt{B.E_{A_2} \times BE_{X_2}}$

$= 78 - \sqrt{36 \times 49}$

$= 78 - 42$

$X_A - X_X = 0.208\sqrt{36}$

$= 0.208 \times 6$

$= 1.248$

$X_X = 3 - 1.248$

$= 1.752$

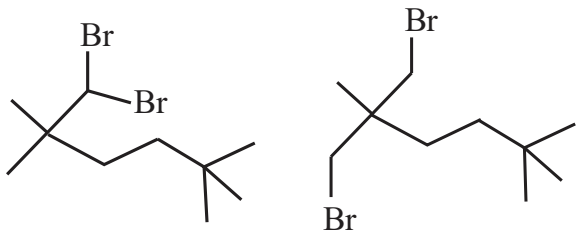
**4.** How many dibromoderivatives of 2, 2, 5, 5 - tetramethyl hexane are there (exluding stereoisomers)

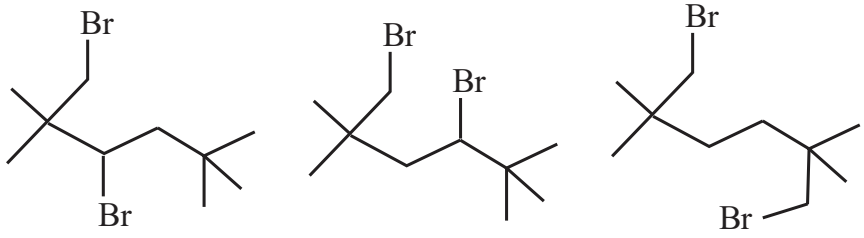
(A) 6 (B) 7

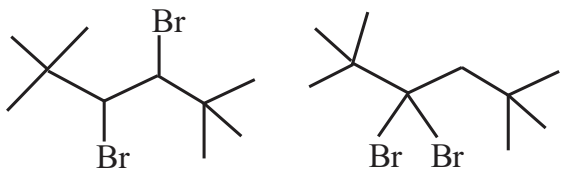
(C) 8 (D) 9

**Ans.** B

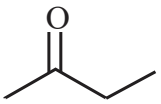
**Sol.**

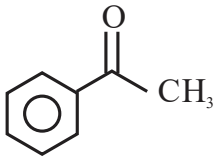


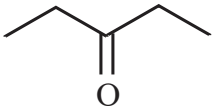


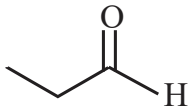


**5.** Which of the following is most reactive toward HCN ?

(A) 

(B) 

(C) 

(D) 

**Ans.** D

**Sol.** D > A > C > B

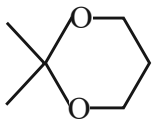
## SECTION–I(ii) : (Maximum Marks : 40)

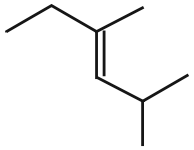
- This section contains **TEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

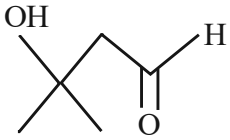
| | | |
|---|---|---|
| *Full Marks* | : +4 | If only (all) the correct option(s) is (are) chosen. |
| *Partial Marks* | : +3 | If all the four options are correct but ONLY three options are chosen. |
| *Partial Marks* | : +2 | If three or more options are correct but ONLY two options are chosen, both of which are correct options. |
| *Partial Marks* | : +1 | If two or more options are correct but ONLY one option is chosen and it is a correct option. |
| *Zero Marks* | : 0 | If none of the options is chosen (i.e. the question is unanswered). |
| *Negative Marks* | : –2 | In all other cases. |

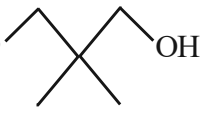
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

**6.** Compound P upon reductive ozonolysis produce Q and R respectively. P upon Hydrogenation with $H_2$/Pd gives 2, 4 - dimethyl hexane. Q undergoes haloform reaction but not Tollen's test. R undergoes crossed aldol reaction with HCHO in presence of dil.NaOH gives only S. S reacts with $NaBH_4/CH_3OH$ gives U. U reacts with $HCHO/H^+$ gives V. Then find correct options ?

(A) V is 

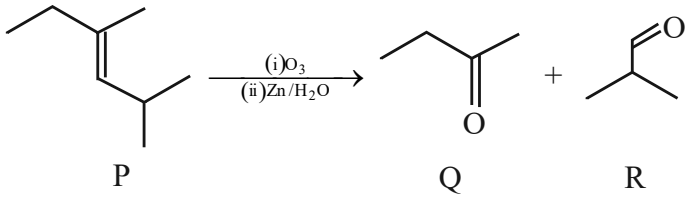
(B) P is 

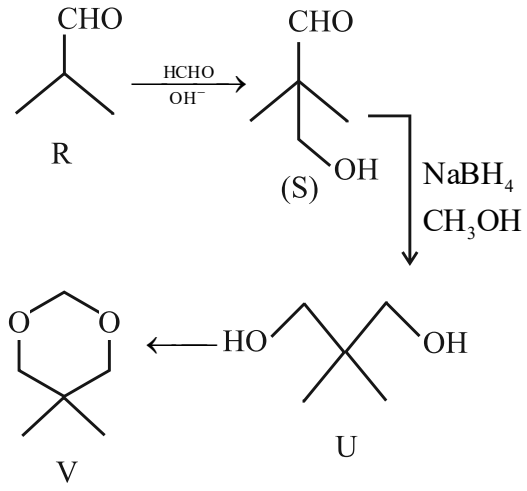
(C) S is 

(D) U is 

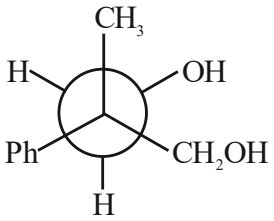
**Ans.** BD

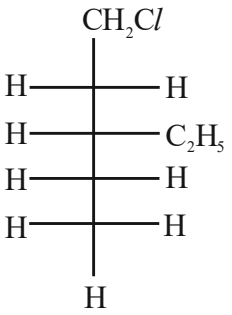
**Sol.**

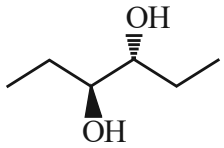


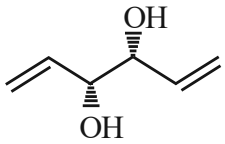




**7.** Among the following compounds, optically inactive is/are

(A) 


(B) 

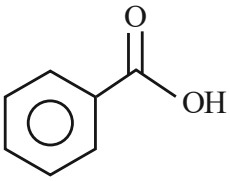

(C) 

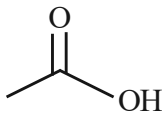

(D) 

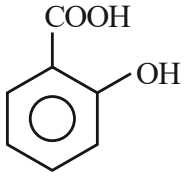

**Ans.** ABC

**Sol.** A, B - do not have chiral carbons

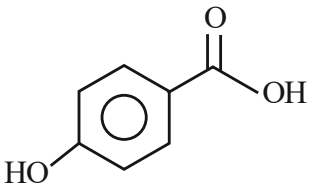
C - is meso compound

**8.** Which of the following more acidic than 


(A) 

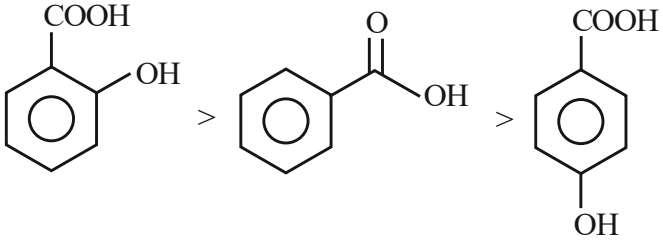

(B) 

(C) HCOOH

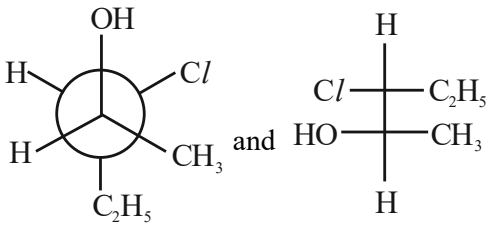
(D) 

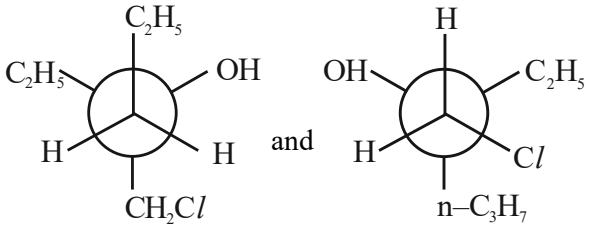
**Ans.** BC

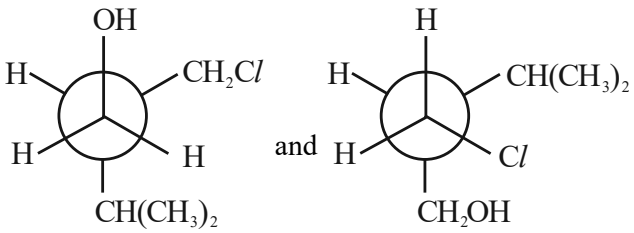
**Sol.** Acidic strength.

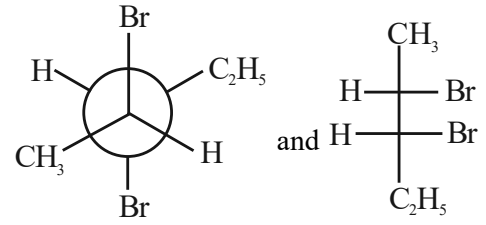


**9.** Which of the following has pair of identical molecules

(A) 

(B) 

(C) 

(D) 

**Ans.** ABCD

**Sol.** (A) Both having R, S configuration

(B) Both having R configuration

(C) Both having S configuration

(D) Both having SR configuration

**10.** Which of the following is/are correctly matched

(A) $Na^{+} > Mg^{+2} > Al^{+3} > O^{-2}$ (radius) (B) $Cr > Mn > V > Sc$ ($I.E_2$)

(C) $Cs_2O > BaO > MgO > Al_2O_3$ [Basic strength] (D) $O > S > Se > Te$ [Electron gain enthalpy]

**Ans.** BC

**Sol.** (A) $O^{-2} > Na^{+} > Mg^{+2} > Al^{+3}$

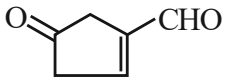
(B) $Cr > Mn > V > Se$

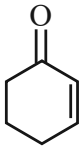
(C) $Cs_2O > BaO > MgO > Al_2O_3$

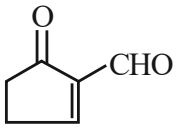
(D) $S > Te > Se > O$

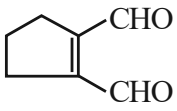
**11.** 


In the above sequence the final products of the reaction is/are :

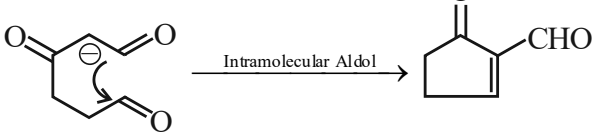
(A) 

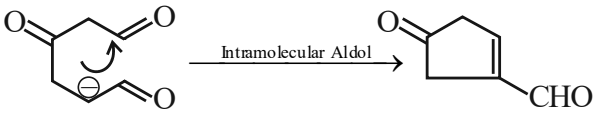

(B) 

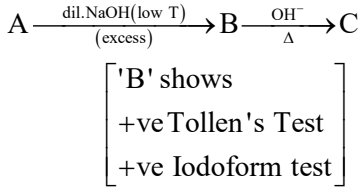

(C) 


(D) 

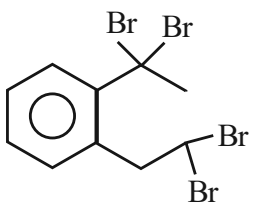

**Ans.** AC

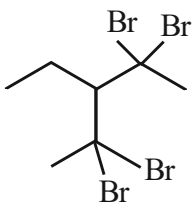
**Sol.** A is 

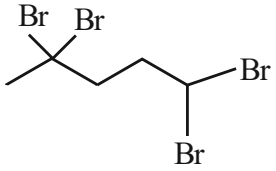

C is 

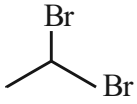

**12.** 


Possible options for A ?

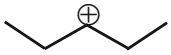
(A) 


(B) 

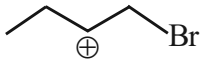
(C) 

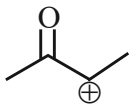
(D) 

**Ans.** ACD

**13.** Which of the following is/are more stable than 

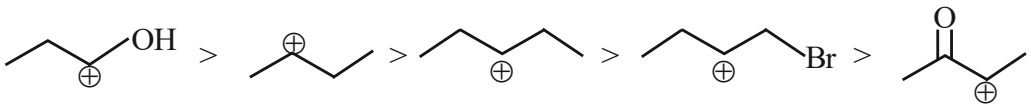
(A) 

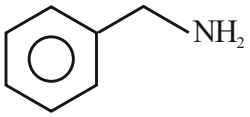
(B) 

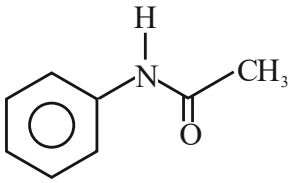
(C) 

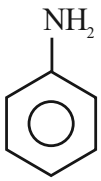
(D) 

**Ans.** AD

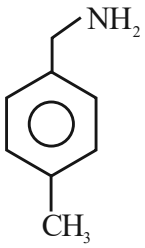
**Sol.** 

**14.** Which of the following is/are less basic than 

(A) 

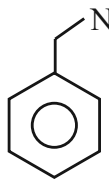
(B) 

(C) 

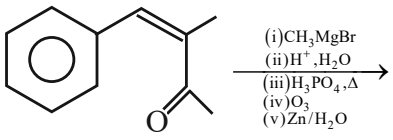
(D) 

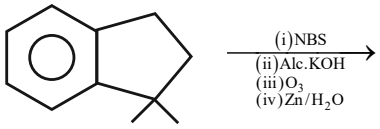
**Ans.** ABC

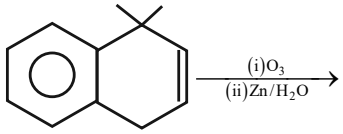
**Sol.** Basic strength

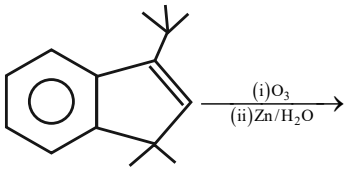
D >  > B > C > A

**15.** In which of the following reaction product gives intramolecular aldol condensation reaction ?

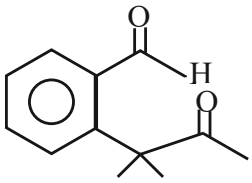
(A) 

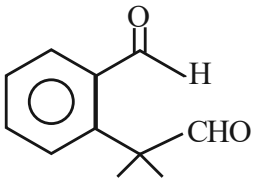

(B) 

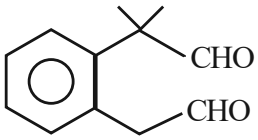
(C) 

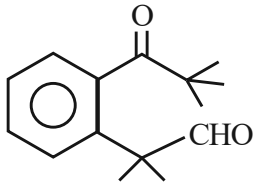

(D) 


**Ans.** AC

**Sol.** A : 

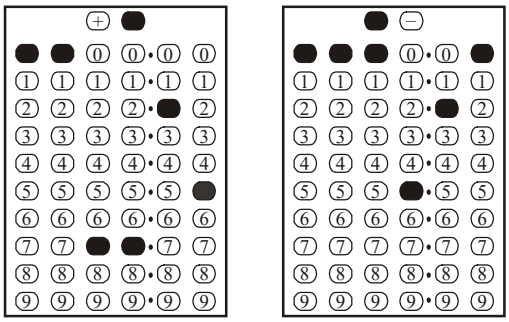

B : 


C : 


D : 

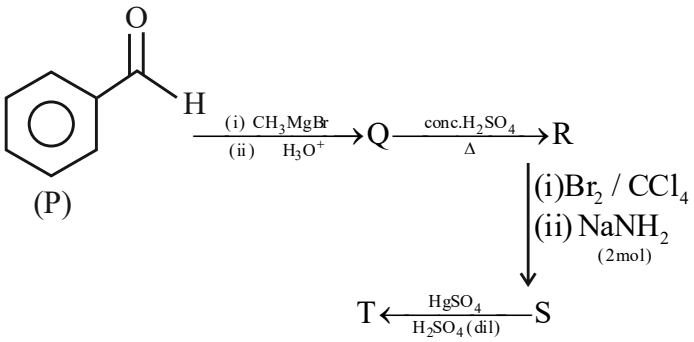

**SECTION-II : (Maximum Marks: 15)**

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
  **For Example :** If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
  *Full Marks* : +3 If ONLY the correct numerical value is entered as answer.
  *Zero Marks* : 0 In all other cases.

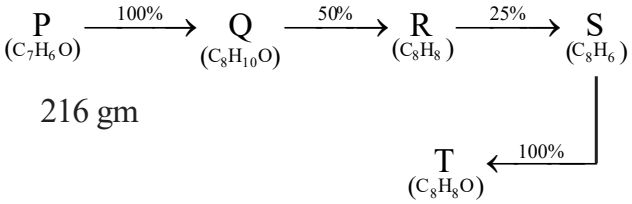
**1.**




Consider the reaction sequence from P to T.
The yield of reaction P to Q 100 %
The yield of reaction Q to R 50%
The yield of reaction R to S 25%
The yield of reaction S to T 100%
What is the amount in grams of T obtained from 216 grams of P.
[Molar mass of C = 12 H = 1 O = 16]

**Ans.** 30

**Sol.**




For 216 gm of P gives 242 gm of Q
242 gm of Q gives 208 gm of R for 100%
so for 50% 104 gm of R
104 gm of R gives 102 gm of S for 100%
so for 25% 25.5 gm of S
25.5 gm of S gives 30 gm of T for 100%

**2.** Ionization potential of element X is 1200 kJ/mole and electron gain enthalpy of element X is –350 kJ/mole. Then find electronegativity of X in

Mulliken Scale $\left[1 \text{ ev / atom} = 96.35 \text{ kJ / mole}\right]$

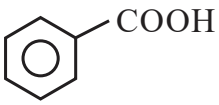
$\{$Answer rounded of to nearest integer$\}$

**Ans.** 8.00

**Sol.** $\text{Electronegativity} = \dfrac{\text{I.P} + \text{E.A}}{2 \times 96.35}$

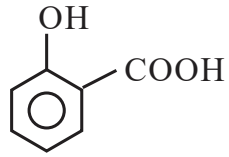
$= \dfrac{1200 + 350}{192.7}$

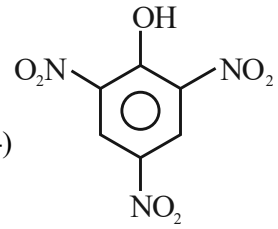
$= 8.04$

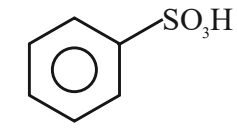
**3.** How many of the following more acidic than 

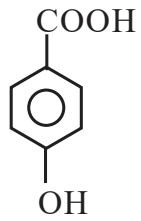
(1) $CH_3COOH$

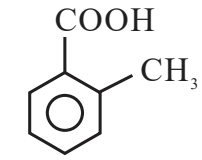
(2) HCOOH

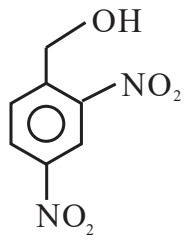
(3) 

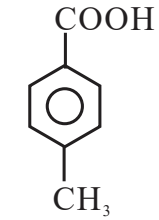
(4) 

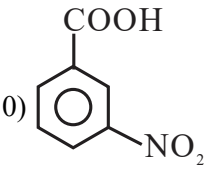
(5) 

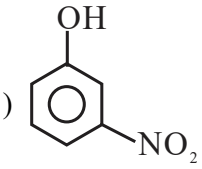
(6) 

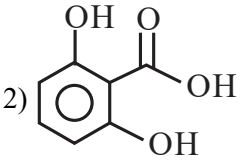
(7) 

(8) 

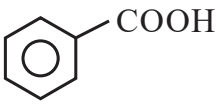
(9) 

(10) 

(11) 

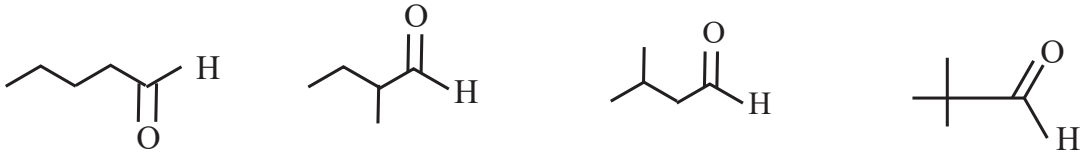
(12) 

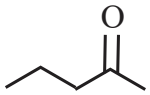
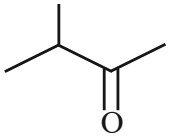
**Ans.** 7

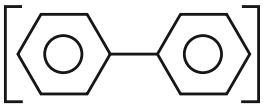
**Sol.** 2, 3, 4, 5, 7, 10, 12 are more acidic than 

**4.** Compound (X) having M.F ($C_5H_{10}O$) [only ketone and aldelyde] reacts with (i) $C_2H_5MgBr$/(ii)$H_3O^+$ gives alcohol. Find no. of X which can give alcohol with at least one chiral carbon.

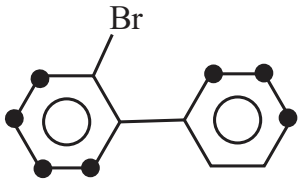
**Ans.** 6

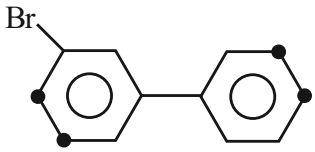
**Sol.** 

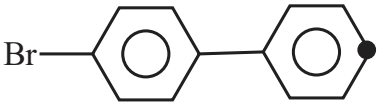



**5.** Number of dibromoderivatives of biphenyl  (only structural isomers)

**Ans.** 12

**Sol.**  : 7

 `: 4

 : 1

= 12

# PART-3 : MATHEMATICS

## SECTION–I(i) : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in <u>one of the following categories</u> :

*Full Marks* : +3 If only the bubble corresponding to the correct option is darkened.
*Zero Marks* : 0 If none of the bubbles is darkened.
*Negative Marks* : –1 In all other cases

---

**1.** The value of the integral $I = \int_2^3 \sqrt{(x-2)(3-x)}dx$ is equal to

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$

(C) $\frac{\pi}{8}$ (D) $2\pi$

**Ans.** C

**Sol.** $I = \int_2^3 \sqrt{(x-2)(3-x)}dx$ put $x = 2\cos^2\theta + 3\sin^2\theta$

$I = \frac{\pi}{8}$

**2.** Let $x > 0$ be a fixed real number. Then the integral $\int_0^{\infty} e^{-t}|x-t|dt$ is equal to

(A) $x+2e^{-x}-1$ (B) $x-2e^{-x}+1$

(C) $x+2e^{-x}+1$ (D) $-x-2e^{-x}+1$

**Ans.** A

**Sol.** $\int_0^{\infty} e^{-t}|x-t|dt = \int_0^{x} e^{-t}(x-t)dt + \int_x^{\infty} e^{-t}.\underbrace{\left(-x+t\right)}_{|}dt$ apply integration by parts

$$=\left[(x-t)\left(-e^{-1}\right)-(-1)e^{-t}\right]_0^x+\left[(x-t)e^{-t}-e^{-t}\right]_x^{\infty} = x+2e^{-x}-1$$

**3.** If $f(x)=xf^2(x)+x^2\ \forall x\in R$, then $\int \frac{f^2(x)-2xf(x)+2x+1}{(1-2xf(x))f(x)\left(f(x)(1-x)+2x+1\right)}dx$ equals

(A) $\frac{-1}{x-f(x)}+c$ (B) $\frac{-1}{x+f(x)}+c$

(C) $\frac{-1}{x^2-f(x)}+c$ (D) $\frac{-1}{x^2+f(x)}+c$

**Ans.** B

**Sol.** $\int \frac{\left(1-2xf(x)+2x+f^2(x)\right)}{(1-2xf(x))\left(x^2+f^2(x)+2xf(x)\right)}dx$

$$=\int \frac{1+\frac{2x+f^2(x)}{1-2xf(x)}}{(x+f(x))^2}dx = \int \frac{1+f'(x)}{(x+f(x))^2}dx = \frac{-1}{x+f(x)}+c$$

**4.** $\int\left(x^{24}+x^{16}+x^8\right)\left(2x^{16}+3x^8+6\right)^{\frac{1}{8}}dx = \frac{\left(2x^{24}+3x^{16}+6x^8\right)^{\frac{\beta}{\gamma}}}{\alpha}+c$ (where $\beta,\gamma$ are co-prime natural numbers and c be constant of integration), then the value of $\alpha+\beta+\gamma$ is

(A) 61 (B) 51

(C) 71 (D) 91

**Ans.** C

**Sol.** $I=\frac{1}{48}\int\left(48x^{23}+48x^{15}+48x^7\right)\left(2x^{24}+3x^{16}+6x^8\right)^{\frac{1}{8}}dx$

Let $2x^{24}+3x^{16}+6x^8=y$

$$I=\frac{1}{48}\int(y)^{\frac{1}{8}}dy=\frac{1}{48}.\frac{8}{9}(y)^{\frac{8}{9}}+c$$

$$I=\frac{1}{54}(y)^{\frac{9}{8}}+c$$

**5.** $\int \frac{(2x+1)}{(x^2+4x+1)^{3/2}}dx$

(A) $\frac{x^3}{(x^2+4x+1)^{1/2}}+C$ (B) $\frac{x}{(x^2+4x+1)^{1/2}}+C$

(C) $\frac{x^2}{(x^2+4x+1)^{1/2}}+C$ (D) $\frac{1}{(x^2+4x+1)^{1/2}}+C$

**Ans.** B

**Sol.** $\int \frac{2x+1}{(x^2+4x+1)^{3/2}}dx=\int \frac{2x+1}{x^3\left(1+\frac{4}{x}+\frac{1}{x^2}\right)^{3/2}}dx$

$\int \frac{2x^{-2}+x^{-3}}{\left(1+\frac{4}{x}+\frac{1}{x^2}\right)^{3/2}}dx$

Now put $\frac{1}{x^2}+\frac{4}{x}+1=t^2$

## SECTION–I(ii) : (Maximum Marks : 40)

- This section contains **TEN** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

| | | |
|---|---|---|
| *Full Marks* | : +4 | If only (all) the correct option(s) is (are) chosen. |
| *Partial Marks* | : +3 | If all the four options are correct but ONLY three options are chosen. |
| *Partial Marks* | : +2 | If three or more options are correct but ONLY two options are chosen, both of which are correct options. |
| *Partial Marks* | : +1 | If two or more options are correct but ONLY one option is chosen and it is a correct option. |
| *Zero Marks* | : 0 | If none of the options is chosen (i.e. the question is unanswered). |
| *Negative Marks* | : –2 | In all other cases. |

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

**6.** Consider an equation, $y\left(3^{|x|}-1\right)+2|x|\left(2^y-1\right)=0$ where $y=\log_2\left(3x^2-1\right)-\log_2\sqrt{x^2+1}$ . Identify which of the following statement(s) is (are) correct ?

(A) Number of real solutions of the equation is 2 (B) Number of real solutions of the equation is 5

(C) Sum of squares of all the solutions is $\frac{7}{9}$ (D) Sum of squares of all the solutions is $\frac{14}{9}$

**Ans.** AD

**7.** $f:[0,1]\to R$ be a function such that $f^2(x)=1$ and $f(x)$ is discontinuous at exactly two point and $\int_0^1 f(x)dx=0$. Let $I=\int_0^1 xf(x)dx$, choose the incorrect option(s)

(A) Maximum value of I is $\frac{1}{4}$

(B) Minimum value of I is $-\frac{3}{4}$

(C) When l is maximum then number of such function f(x) is 8

(D) When l is minimum then number of such function f(x) is 4

**Ans.** BCD

**Sol.** Let f(x) = $\begin{matrix}1 & 0\le x<a\\ -1 & a\le x<b\\ 1 & b\le x<1\end{matrix}$

As $\int_0^1 f(x)dx=0\Rightarrow a-(b-a)+(1-b)=0;2a+1=2b$

$$\int_0^1 xf(x)=\int_0^a x\,fdx+\int_a^b xf(x)dx+\int_b^1 xf(x)dx=\frac{a^2}{2}-\left(\frac{b^2-a^2}{2}\right)+\frac{1-b^2}{2}=\frac{2a^2-2b^2+1}{2}$$

$$l=\frac{2a^2-2\left(\frac{2a+1}{2}\right)^2+1}{2}=\frac{1-4a}{4};I_{max}=\frac{1}{4}\text{ similarly minimum is }-\frac{1}{4}$$

**8.** If $\int_{-\alpha}^{\alpha}\left\{e^x+(\cos x)\left(\ell n\left(x+\sqrt{1+x^2}\right)\right)\right\}dx>\frac{3}{2}$, then $\alpha$ can be

(A) 1 (B) 2

(C) 3 (D) 4

**Ans.** ABCD

**9.** If $\int_0^1\left\{\left(1-x^{2018}\right)^{\frac{1}{2020}}-\left(1-x^{2020}\right)^{\frac{1}{2018}}\right\}.dx=J$, then

(A) $J<\frac{2018}{2020}$ (B) $J>\frac{2018}{2020}$

(C) $J>\frac{2020}{2018}$ (D) $J<\frac{2020}{2018}$

**Ans.** AD

**Sol.** Let $I_1=\int_0^1\left\{\left(1-x^{2018}\right)^{\frac{1}{2020}}\right\}.1.dx$, now use I.B.P.

$$= \left[\left(1-x^{2018}\right)^{\frac{1}{2020}}.x\right]_0^1 + \frac{2018}{2020}\int_0^1 \frac{x^{2017}}{\left(1-x^{2018}\right)^{\frac{2019}{2020}}}.x.dx$$

now put $1-x^{2018} = t^{2020}$ in integration

$$= 0 + \int_0^1 \left(1-t^{2020}\right)^{\frac{1}{2018}}.dt = I_2\,(\text{let}),\ \text{now } J = I_1 - I_2 = 0$$

**10.** If $\int\left(2x^6+15x^4+2x^2+3\right)\cos(2x)dx = f(x)\sin(2x)+g(x)\cos(2x)+K$, where f(x) and g (x) are polynomial of x such that f (0) = 1, g (–1) = –4 and K is constant of

(A) $4f(1) \le 3g(1)$

(B) $\int_{-\pi/3}^{\pi/3} f(x)dx \neq 0$

(C) $g''(0) < 10$

(D) $\frac{g(2)}{f(2)} > 1$

**Ans.** ABCD

**11.** Which of the following function(s) is/are even ?

(A) $f(x) = \int_0^x \ln\left(t+\sqrt{1+t^2}\right)dt$

(B) $g(x) = \int_0^x \frac{\left(2^t+1\right)t}{2^t-1}dt$

(C) $h(x) = \int_0^x \left(\sqrt{1+t+t^2}-\sqrt{1-t+t^2}\right)dt$

(D) $\ell(x) = \int_0^x \ln\left(\frac{1-t}{1+t}\right)dt$

**Ans.** ACD

**Sol.** By Leibniz's rule

**12.** If $\int \frac{3x\sin^2 x\cos x - 3\sin^3 x}{x^4}dx = f(x)+c$, $\lim_{x\to 0} f(x) = 1$, then which of the following is/are true.

(A) $\lim_{x\to 0}\frac{\int_0^x tf(t)dt - 2x^2}{1-\cos x} = -3$

(B) $\lim_{x\to 0}\frac{\left(f(x)\right)^{\frac{1}{3}} - x^2}{x^2} = -\frac{1}{6}$

(C) If $h(x) = x.\sqrt[3]{f(x)}$, then $\int_0^{\pi} h^4(x)dx = \frac{3\pi}{8}$

(D) If $h(x) = x.\sqrt[3]{f(x)}$, then $\int_0^{\pi/2} e^{h(x)}\left(\cos^2 x - \sin x\right)dx = -1$

**Ans.** ACD

**Sol.** $\int \frac{3\sin^2 x\cos x}{x^3} - 3\int \frac{\sin^3 x}{x^4}dx$

$$\int \frac{3\sin^2 x\cos x}{x^3}-3\sin^3 x.\left(\frac{x^{-3}}{-3}\right)+\int 3\sin^2 x\cos x\left(\frac{-1}{x^3}\right)dx$$

$$\therefore f(x)=\frac{\sin^3 x}{x^3}$$

**13.** If $f:(0,1)\to R$ is defined by $f(x)=\sum_{n=0}^{\infty}\frac{x^{2^n}}{1-x^{2^{n+1}}}$, then –

(A) $f^{-1}(x)=\frac{x}{x+1}$

(B) $\int_0^{1/2} f(x)\,dx=\frac{2\ell n2-1}{2}$

(C) $\int_0^1 f(x)\,dx=\ell n2-1$

(D) $\int_0^1 f(x)\,dx=\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+.....\infty$

**Ans.** BD

**14.** Let $I=\int_0^1\sqrt{\frac{1+\sqrt{x}}{1-\sqrt{x}}}\,dx$ and $J=\int_0^1\sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}}\,dx$ then which statement(s) are correct

(A) $I+J=2$

(B) $I-J=\pi$

(C) $J=\frac{2+\pi}{2}$

(D) $J=\frac{4-\pi}{2}$

**Ans.** BD

**Sol.** Put $x=\sin^2\theta$

**15.** $\int\frac{x^4+1}{x^6+1}dx$ is

(A) $\tan^{-1}x+\frac{1}{3}\tan^{-1}(x^3)+c$

(B) $\frac{1}{3}\tan^{-1}\left(\frac{3x-3x^5}{1-3x^2-3x^4+x^6}\right)+c$

(C) $\frac{1}{3}\tan^{-1}\left(\frac{2x-x^4}{1-2x^5-3x^2}\right)+c$

(D) $\frac{1}{2}\tan^{-1}\left(\frac{x-x^3-1}{1+x^2-3x^5}\right)+c$

**Ans.** AB

**Sol.** $\int\frac{x^4+1}{x^6+1}dx=\int\frac{x^4-x^2+1}{x^6+1}dx+\int\frac{x^2}{x^6+1}dx$

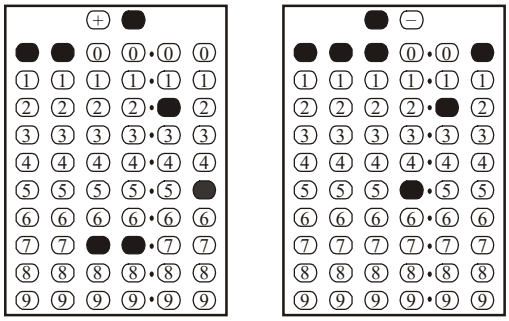
$$=\int\frac{(x^4-x^2+1)}{(x^2+1)(x^4-x^2+1)}dx+\int\frac{x^2}{(x^3)^2+1}dx$$

$$=\tan^{-1}x+\frac{1}{3}\tan^{-1}x^3$$

$$=\frac{1}{3}\tan^{-1}\left(\frac{3x-3x^5}{1-3x^2-3x^4+x^6}\right)+c$$

**SECTION-II : (Maximum Marks: 15)**

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, –0.33, –.30, 30.27, –127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.
  **For Example :** If answer is –77.25, 5.2 then fill the bubbles as follows.



- Answer to each question will be evaluated according to the following marking scheme:
  *Full Marks* : +3 If ONLY the correct numerical value is entered as answer.
  *Zero Marks* : 0 In all other cases.

---

**1.** Let $f(1)+g(1)=9e; f(x)=-x^2g'(x)$ and $g(x)=-x^2f'(x)$. If

$$\int_1^4\left(\frac{f(x)+g(x)}{x^2}\right)dx=k\left(e-e^{\frac{1}{4}}\right), \text{then}\left|\frac{k}{2}\right| \text{ is}$$

**Ans.** 4.5

**Sol.** $$1=\int_1^4\frac{-x^2\left(f'(x)+g'(x)\right)}{x^2}dx=-\left[f(x)+g(x)\right]_1^4=-\left(f(4)+g(4)-9e\right)$$

Now, $f(x)+g(x)=-x^2\left(f'(x)+g('(x)\right)$

$$\Rightarrow\int\frac{f'(x)+g'(x)}{f(x)+g(x)}dx=-\int x^{-2}dx=\frac{1}{x}+c \Rightarrow f(x)+g(x)=ae^{\frac{1}{x}} \Rightarrow a=9$$

$$1=-\left(9e^{\frac{1}{4}}-9e\right)=9\left(e-e^{\frac{1}{4}}\right)$$

**2.** If $f\left(x+f(y)\right)+f\left(x-f(y)\right)=f(x)\forall\, x,y\in R$ and (0, 2) lies on $y=f(x)$, then

$$\int_0^{12}f(x)dx=\lambda\int_{-24}^{24}f(x-36)dx \text{ where } \lambda \text{ equals}$$

**Ans.** 0.25

**Sol.** $f(0)=2\Rightarrow f(x+2)+f(x-2)=f(x)\Rightarrow f(x)=f(x+12)$

$$\int_0^{12}f(x)dx=\frac{1}{4}\int_{-24}^{24}f(x-36)dx$$

**3.** $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{2022-2023\cos ec^2 x}{\sin^{2022} x}dx$ equals $-\frac{2^p\left(3^q-1\right)}{3^r}$, then p + q + r is equal to

**Ans.** 4045.50

**Sol.** $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{2022-2023\cos ec^2 x}{\sin^{2022} x}dx = \left|\frac{\cot x}{\sin^{2022} x}\right|_{\frac{\pi}{6}}^{\frac{\pi}{3}} = -\frac{2^{2022}\left(3^{1012}-1\right)}{3^{1011.5}}$

**4.** If $\int \frac{\left(\sqrt{x}\right)^5}{\left(\sqrt{x}\right)^7+x^6}dx = a.\ln\left(\frac{x^{k/2}}{x^{k/2}+1}\right)+C$, then the value of k is (given a > 0)

**Ans.** 5

**Sol.** Put $I = 2\int \frac{dt}{t^6\left(1+\frac{1}{t^5}\right)}$, put $1+\frac{1}{t^5} = y \Rightarrow I = \frac{2}{5}\ln\left(\frac{x^{5/2}}{x^{5/2}+1}\right)+C$

$\therefore\ a = \frac{2}{5}$ and $k = 5$

**5.** If $A = \{a_1, a_2, .....a_n\}$ be set of all possible natural factors of 1000 and $P = \int_{-2}^{2} x\ln\left(a_1^x + a_2^x + ..... + a_n^x\right)dx$,

then value of $\frac{2P}{n}$ equals lnk, find k.

**Ans.** 10

**Sol.** $1000 = 5^3 . 2^3$

Number of factors (3+1) (3+1) =16

Applying king , $P = -\int_{-2}^{2} x\ln\left(a_1^{-x} + a_2^{-x} + ..... + a_{16}^{-x}\right)dx$

$= -\int_{-2}^{2} x\ln\left(\frac{1}{a_1^x}+\frac{1}{a_2^x}+.....+\frac{1}{a_{16}^x}\right)dx$

$= -\int_{-2}^{2} x\ln\left(\frac{a_1^x+a_2^x+.....+a_{16}^x}{1000^x}\right)dx$

$= -\int_{-2}^{2} x\ln\left(a_1^x+a_2^x+.....+a_{16}^x\right)dx + \int_{-2}^{2} x^2\ln(1000)dx$

$2P = \frac{16}{3}\ln 1000 \Rightarrow p = 8\ln 10 \Rightarrow \frac{2P}{n} = \ln 10$